Sample: site and customer names removed · www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

Single-line diagrams, EV charging stations

Electrical design basis for the four stations · cable, transformer and accessory specs · BOM to follow

Site	Power	Transformer	Main breaker	Charger breakers	Charger cable (per charger)
Amphoe Mueang Lamphun	240 kW	315 kVA	MCCB 3P 630 AF / 500 AT	MCCB 3P 250 AF / 250 AT × 2	CV 4 × 1C 120 mm²
Market car park, Mukdahan	360 kW	500 kVA	ACB 3P 800 AF / 800 AT	MCCB 3P 250 AF / 250 AT × 3	CV 4 × 1C 120 mm²
Community mall, Ubon Ratchathani	360 kW	500 kVA	ACB 3P 800 AF / 800 AT	MCCB 3P 160 AF / 160 AT × 4	CV 4 × 1C 70 mm²
Roadside lot, Nakhon Ratchasima	540 kW	630 kVA	ACB 3P 1,000 AF / 1,000 AT	MCCB 3P 400 AF / 400 AT × 3	CV 4 × 1C 185 mm²

Cables

- LV power: CV (XLPE/PVC) 0.6/1 kV, Cu, TIS 2143, single-core
- Earth: THW Cu green/yellow, TIS 11
- HV: XLPE 24 kV Cu, screened, TIS 2202
- Sizes allow for grouping in ducts (0.8) and 40 °C ground
- Voltage drop ≤ 3 % from MDB to each charger at full load

Switchgear

- ACB/MCCB to IEC 60947-2, Icu ≥ the fault level shown
- Charger MCCBs with shunt trip, wired to the island emergency stop
- MDB outdoor IP54, Form 2b, with fans and filtered vents
- Busbar Cu, phase colours brown/black/grey, N blue, PE green/yellow
- Lockable main breaker for safe isolation

Protection and metering

- SPD Type 1+2 at MDB, Type 2 at DB-AUX
- Each charger has internal RCD and SPD (per datasheet)
- PEA TOU meter on CT, position agreed with PEA
- Multifunction meter with Modbus for remote monitoring
- Phase-loss / under-voltage relay on the main breaker (recommended)

Earthing and testing

- TN-S from the transformer; neutral and PE separate after the MDB
- Earth resistance ≤ 5 Ω, measured and recorded
- Insulation test ≥ 1 MΩ at 1,000 V DC on every cable
- Torque check of all terminations, thermal scan at first full load
- Design and as-built SLD signed by a licensed electrical engineer

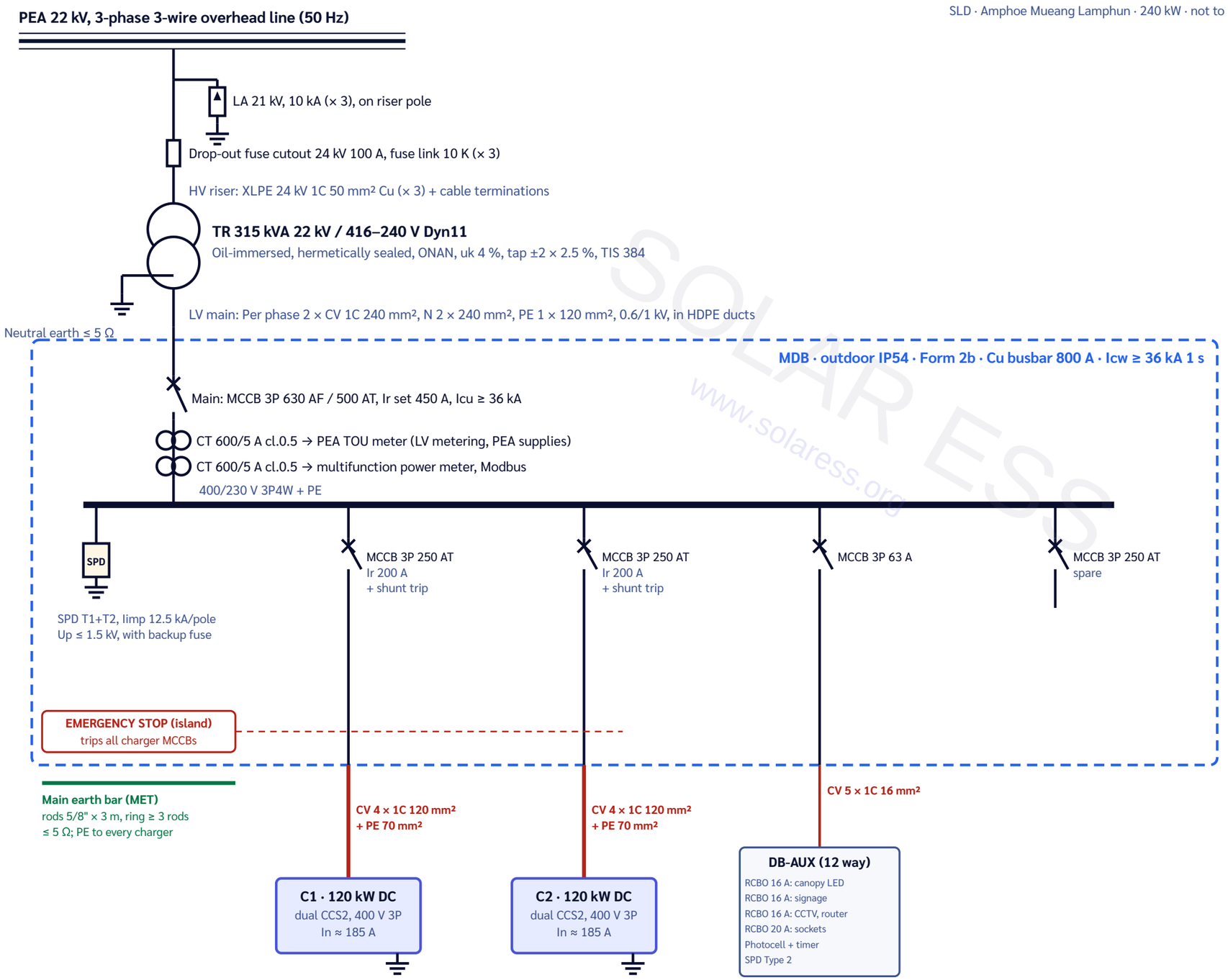
Indicative design from the project decks and charger datasheets. Final sizes depend on site survey, route lengths, PEA requirements and the chosen charger model.

SOLAR ESS www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

SLD 1. Amphoe Mueang Lamphun

Standalone charging hub · Highway 114 · 240 kW · 2 × 120 kW



Specification

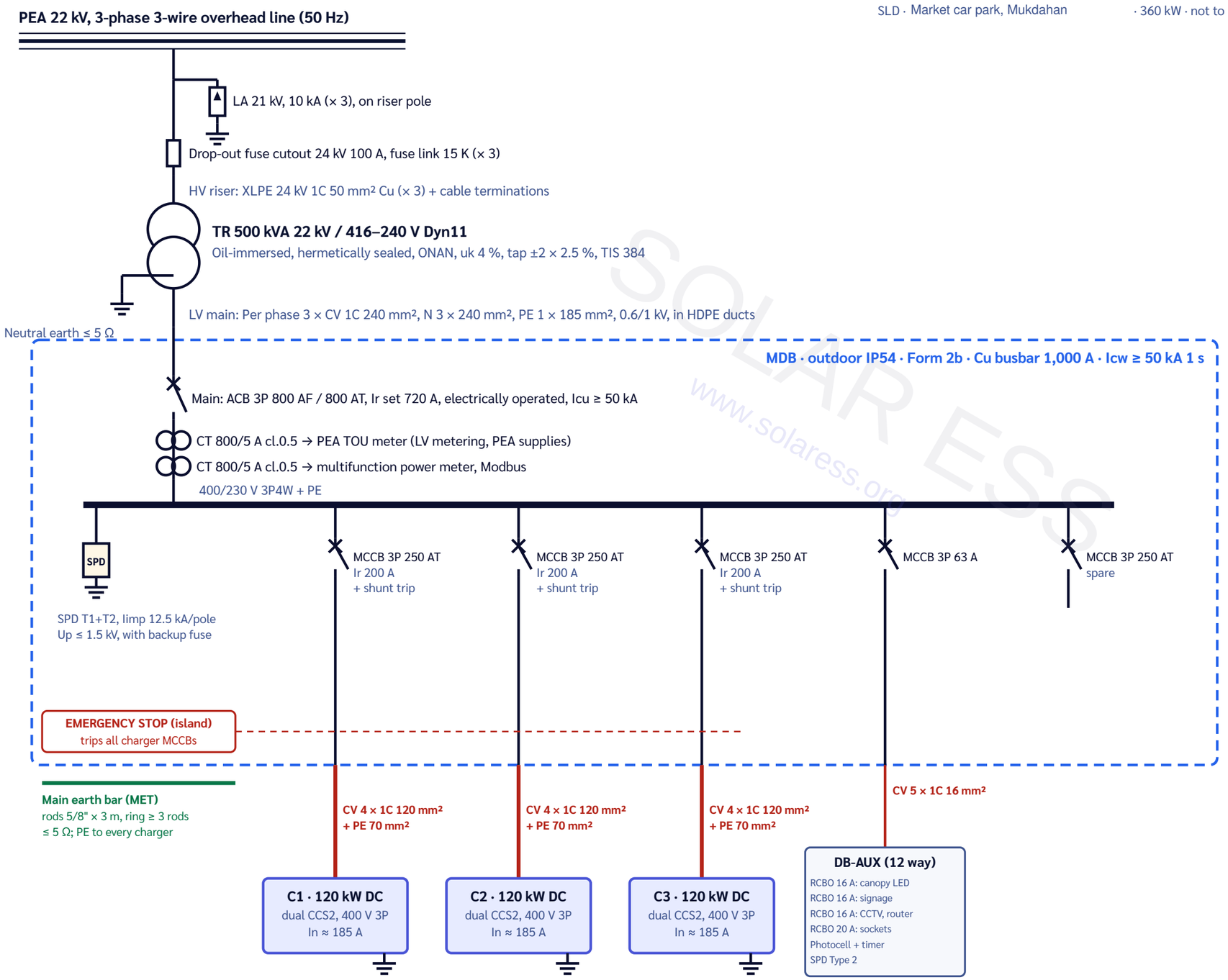
HV supply	PEA 22 kV 3-phase overhead, riser pole near the site
Lightning arrester	21 kV, 10 kA, polymer, 3 pcs, earthed ≤ 5 Ω
HV fuse	Drop-out fuse cutout 24 kV 100 A, fuse link 10 K (per PEA table)
HV cable	XLPE 24 kV 1C 50 mm² Cu, 3 pcs, with outdoor terminations
Transformer	315 kVA, 22 kV ±2 × 2.5 % / 416–240 V, Dyn11, uk 4 %, oil, hermetically sealed, TIS 384, PEA approved list
LV full-load current	437 A; LV fault level ≈ 11 kA
Load	2 × 120 kW ≈ 370 A at full output (85 % of transformer)
Transformer → MDB	Per phase 2 × CV 1C 240 mm², N 2 × 240 mm², PE 1 × 120 mm², CV 0.6/1 kV, in HDPE ducts
MDB main breaker	MCCB 3P 630 AF / 500 AT, Ir set 450 A, Icu ≥ 36 kA
Metering	PEA TOU meter with CT 600/5 A cl.0.5; multifunction meter (V, A, kW, kWh, THD) with own CTs
Surge protection	MDB: SPD Type 1+2, 12.5 kA/pole; DB-AUX: Type 2; chargers: built-in
Charger feeder breaker	MCCB 3P 250 AF / 250 AT, Ir 200 A, shunt trip to emergency stop, Icu ≥ 36 kA
Charger cable	CV 4 × 1C 120 mm² + THW 1C 70 mm² (G/Y), CV 0.6/1 kV Cu, in HDPE Ø125
Aux DB	MCCB 3P 63 A feeder, CV 5 × 1C 16 mm²; RCBO 30 mA per circuit
Earthing	TN-S; rods 5/8" × 3 m, ring ≤ 5 Ω; transformer, MDB, canopy and chargers bonded

Cable sizes include 0.8 derating for grouped ducts at 40 °C; check voltage drop ≤ 3 % on the final route lengths.

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

SLD 2. Market car park, Mukdahan

Market car park · next to fuel station · 360 kW · 3 × 120 kW



Specification

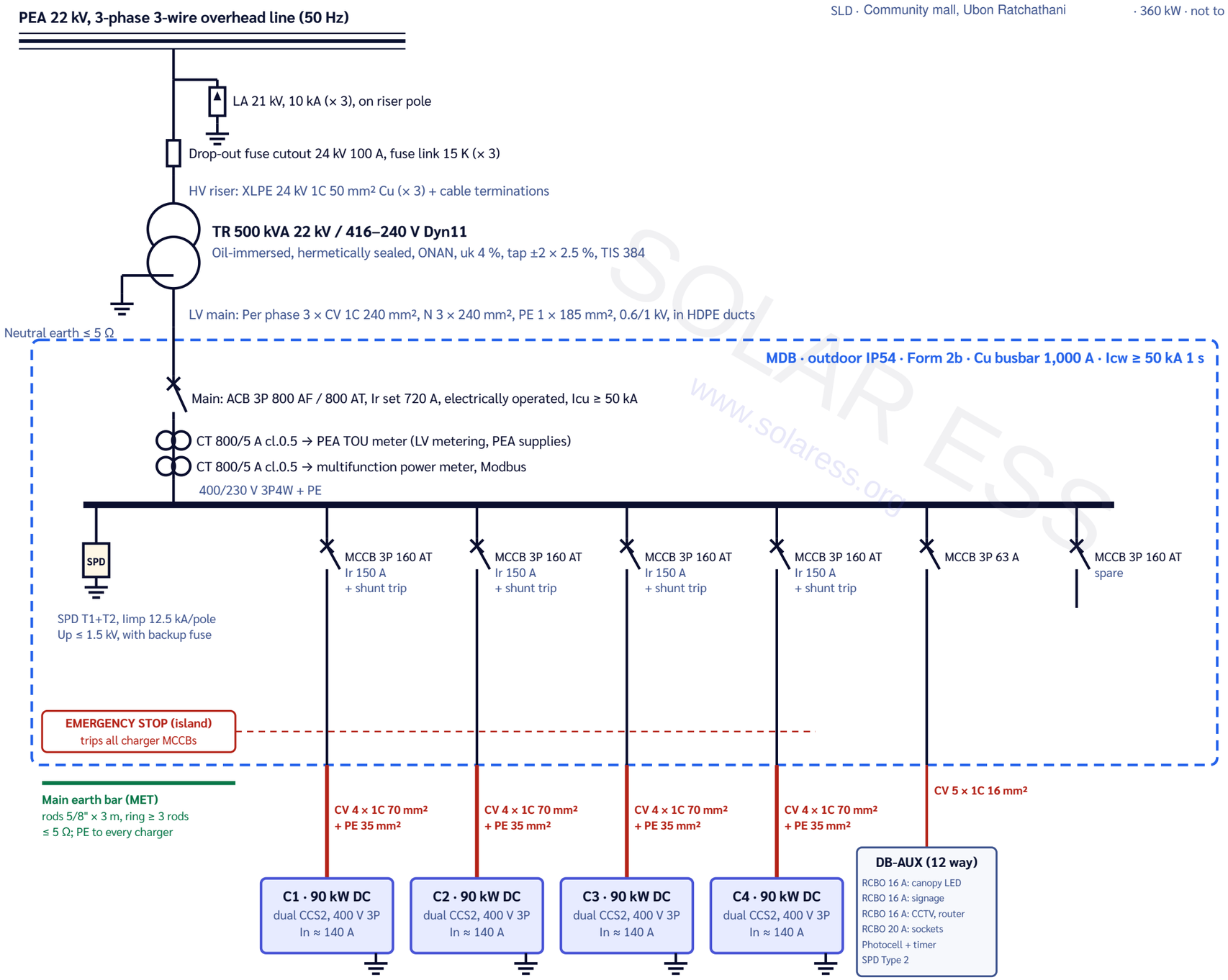
HV supply	PEA 22 kV 3-phase overhead, riser pole near the site
Lightning arrester	21 kV, 10 kA, polymer, 3 pcs, earthed ≤ 5 Ω
HV fuse	Drop-out fuse cutout 24 kV 100 A, fuse link 15 K (per PEA table)
HV cable	XLPE 24 kV 1C 50 mm² Cu, 3 pcs, with outdoor terminations
Transformer	500 kVA, 22 kV ±2 × 2.5 % / 416–240 V, Dyn11, uk 4 %, oil, hermetically sealed, TIS 384, PEA approved list
LV full-load current	694 A; LV fault level ≈ 18 kA
Load	3 × 120 kW ≈ 555 A at full output (80 % of transformer)
Transformer → MDB	Per phase 3 × CV 1C 240 mm², N 3 × 240 mm², PE 1 × 185 mm², CV 0.6/1 kV, in HDPE ducts
MDB main breaker	ACB 3P 800 AF / 800 AT, Ir set 720 A, electrically operated, Icu ≥ 50 kA
Metering	PEA TOU meter with CT 800/5 A cl.0.5; multifunction meter (V, A, kW, kWh, THD) with own CTs
Surge protection	MDB: SPD Type 1+2, 12.5 kA/pole; DB-AUX: Type 2; chargers: built-in
Charger feeder breaker	MCCB 3P 250 AF / 250 AT, Ir 200 A, shunt trip to emergency stop, Icu ≥ 50 kA
Charger cable	CV 4 × 1C 120 mm² + THW 1C 70 mm² (G/Y), CV 0.6/1 kV Cu, in HDPE Ø125
Aux DB	MCCB 3P 63 A feeder, CV 5 × 1C 16 mm²; RCBO 30 mA per circuit
Earthing	TN-S; rods 5/8" × 3 m, ring ≤ 5 Ω; transformer, MDB, canopy and chargers bonded

Cable sizes include 0.8 derating for grouped ducts at 40 °C; check voltage drop ≤ 3 % on the final route lengths.

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

SLD 3. Community mall, Ubon Ratchathani

Community mall car park · 360 kW · 4 × 90 kW



Specification

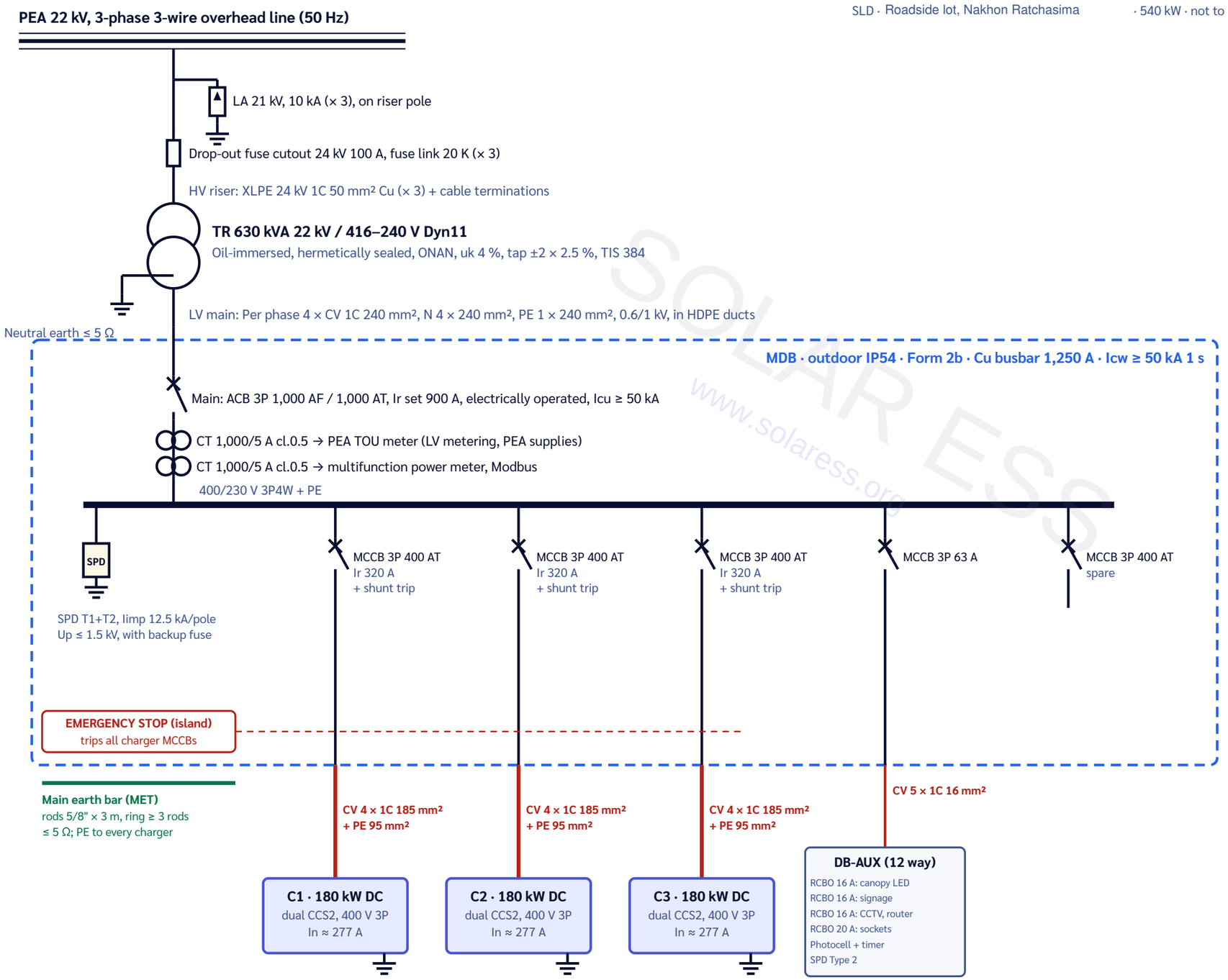
HV supply	PEA 22 kV 3-phase overhead, riser pole near the site
Lightning arrester	21 kV, 10 kA, polymer, 3 pcs, earthed ≤ 5 Ω
HV fuse	Drop-out fuse cutout 24 kV 100 A, fuse link 15 K (per PEA table)
HV cable	XLPE 24 kV 1C 50 mm² Cu, 3 pcs, with outdoor terminations
Transformer	500 kVA, 22 kV ±2 × 2.5 % / 416–240 V, Dyn11, uk 4 %, oil, hermetically sealed, TIS 384, PEA approved list
LV full-load current	694 A; LV fault level ≈ 18 kA
Load	4 × 90 kW ≈ 560 A at full output (81 % of transformer)
Transformer → MDB	Per phase 3 × CV 1C 240 mm², N 3 × 240 mm², PE 1 × 185 mm², CV 0.6/1 kV, in HDPE ducts
MDB main breaker	ACB 3P 800 AF / 800 AT, Ir set 720 A, electrically operated, Icu ≥ 50 kA
Metering	PEA TOU meter with CT 800/5 A cl.0.5; multifunction meter (V, A, kW, kWh, THD) with own CTs
Surge protection	MDB: SPD Type 1+2, 12.5 kA/pole; DB-AUX: Type 2; chargers: built-in
Charger feeder breaker	MCCB 3P 160 AF / 160 AT, Ir 150 A, shunt trip to emergency stop, Icu ≥ 50 kA
Charger cable	CV 4 × 1C 70 mm² + THW 1C 35 mm² (G/Y), CV 0.6/1 kV Cu, in HDPE Ø110
Aux DB	MCCB 3P 63 A feeder, CV 5 × 1C 16 mm²; RCBO 30 mA per circuit
Earthing	TN-S; rods 5/8" × 3 m, ring ≤ 5 Ω; transformer, MDB, canopy and chargers bonded

Charger split assumed 4 × 90 kW. If 2 × 180 kW is chosen, use the 180 kW feeder (MCCB 400 A, 185 mm²).

Solar ESS Co. Ltd.
53 Moo 2, Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

SLD 4. Roadside lot, Nakhon Ratchasima

Near a university · empty lot · 540 kW · 3 × 180 kW



Specification

HV supply	PEA 22 kV 3-phase overhead, riser pole near the site
Lightning arrester	21 kV, 10 kA, polymer, 3 pcs, earthed ≤ 5 Ω
HV fuse	Drop-out fuse cutout 24 kV 100 A, fuse link 20 K (per PEA table)
HV cable	XLPE 24 kV 1C 50 mm² Cu, 3 pcs, with outdoor terminations
Transformer	630 kVA, 22 kV ±2 × 2.5 % / 416–240 V, Dyn11, uk 4 %, oil, hermetically sealed, TIS 384, PEA approved list
LV full-load current	874 A; LV fault level ≈ 22 kA
Load	3 × 180 kW ≈ 831 A at full output (95 % of transformer)
Transformer → MDB	Per phase 4 × CV 1C 240 mm², N 4 × 240 mm², PE 1 × 240 mm², CV 0.6/1 kV, in HDPE ducts
MDB main breaker	ACB 3P 1,000 AF / 1,000 AT, Ir set 900 A, electrically operated, Icu ≥ 50 kA
Metering	PEA TOU meter with CT 1,000/5 A cl.0.5; multifunction meter (V, A, kW, kWh, THD) with own CTs
Surge protection	MDB: SPD Type 1+2, 12.5 kA/pole; DB-AUX: Type 2; chargers: built-in
Charger feeder breaker	MCCB 3P 400 AF / 400 AT, Ir 320 A, shunt trip to emergency stop, Icu ≥ 50 kA
Charger cable	CV 4 × 1C 185 mm² + THW 1C 95 mm² (G/Y), CV 0.6/1 kV Cu, in HDPE Ø160
Aux DB	MCCB 3P 63 A feeder, CV 5 × 1C 16 mm²; RCBO 30 mA per circuit
Earthing	TN-S; rods 5/8" × 3 m, ring ≤ 5 Ω; transformer, MDB, canopy and chargers bonded

> Transformer loading is high (≈ 95 % at full output). Use the chargers' dynamic power limit, or upsize to 800 kVA if PEA allows.